Google Developer Group
Bandung Institute of Technology

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| **Team name** | Tide Ledger |
| **Project title** | RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports |
| **Selected case** | Case 1: Smarter Campus and Learning |
| **Members** | Vincentius Bryan Kwandou (leader), Valentino Filemon Limer, Putra Purwanugraha Tengbunan, Marvel Harjosetio |


## Abstract

**1. Problem.** Student treasurers manage millions of rupiah without accounting training under complex faculty bylaws: physical stamps, arithmetic balance, attendance ratios, revenue stamps on receipts exceeding Rp5,000,000 under Law 10/2020, and tight deadlines. At Brawijaya Computer Science, incomplete reports halt upcoming funding disbursements and extracurricular learning [6]. Treasurers lose valuable study hours to mechanical corrections, turning accountability into anxiety rather than educational governance.

**2. Intended Users.** Primary users: student organization treasurers and secretaries across Indonesian universities (BEM and clubs) lacking accounting backgrounds. Secondary users: faculty student affairs verifiers (Biro Kemahasiswaan and student council auditors) who conduct pre-disbursement desk reviews.

**3. Supporting Observations or Evidence.** Faculty research at Universitas Atma Jaya Makassar documented that the Student Affairs Bureau (BKAM) faces unstandardized event portfolio records and manual tracking that overburden student organizations [1]. Disparate faculty guidelines compound confusion: UB mandates stamped receipts and ledger balances [2]; UGM requires Rp10,000 stamps on receipts exceeding Rp5,000,000 under Law 10/2020 and photocopies for thermal slips [3]; ITS enforces 10-day deadlines [4] and tax inclusion [5]. Student officers consistently struggle with tax deductions and budget reconciliations [6].

**4. Proposed Solution and Core User Flow.** RapiLPJ transforms administrative triage into experiential fiscal learning across three steps: (1) Multimodal Intake & Deterministic Audit: Gemini 2.0 Flash extracts structured fiscal metadata from receipt photos into JSON; a client-side JavaScript engine audits zero-tolerance rules (duplicate invoices, date bounds, >Rp5M revenue stamps, >10% budget variances) without calculation hallucinations. (2) Fiscal Literacy Loop & Addenda: For every infraction, an interactive module presents statutory rationale (Law 10/2020, archival durability) with remediation micro-quizzes and synthesizes formal budget variance addenda. (3) Verifier Sign-off & PDF Dossier: Treasurers export an indexed receipt PDF bound by a SHA-256 hash, verified in seconds by faculty reviewers.

**5. How It Differs from Existing Alternatives.** Pure LLMs hallucinate calculations on balance sheets. Spreadsheets lack automated statutory checks, receipt image indexing, and educational remediation loops. Campus portals merely collect finished files without pre-submission triage. RapiLPJ enforces a strict neuro-symbolic boundary: Gemini 2.0 Flash handles visual perception and governance guidance, while financial arithmetic and compliance verification are 100% reserved for transparent, deterministic client-side code.

**6. Planned Technologies.** A privacy-preserving client-side stack. Perception & Guidance: Gemini 2.0 Flash Multimodal API (client-side BYOK) for receipt extraction and statutory drafting. Deterministic Kernel: Open-source JavaScript rule evaluator, FileReader receipt compiler, portable .lpj JSON state persistence, and SHA-256 cryptographic hashing. Presentation: Accessible HTML5/CSS3 (bilingual EN/ID and light/dark themes) deployed statically on Vercel CDN without database storage, ensuring student financial confidentiality.

**7. Expected Benefit and Development Plan.** Expected benefit: targets reducing mechanical administrative errors by >80% prior to desk verification, compressing review turnarounds from a 7–14 day baseline to under 1 business day, and building measurable fiscal literacy via remediation micro-quizzes. Development plan: Semifinal phase pilots the engine with 30 treasurers and 2 faculty advisors at Universitas Atma Jaya Makassar, standardizing the primary institutional preset; Final Day sprint delivers multi-page batch OCR scanning and QR-verifiable advisor sign-off. Modular policy toggles mitigate guideline changes.



**Source Acknowledgments.** [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026. AI disclosure: Claude and Gemini supported research, deterministic engine code, and drafting.